AGI-IO Creator Application

The agi-io_createor is just a standard Python program and should run on any platform with Python 3 installed.

There are no major dependencies to running the application other than NUMPY and the TKINTER libraries (see the requirements.txt for the specific version numbers).

Launching the Application:

Name	Type	Compressed size
agi-io_creator.py	Python File	14 KB
CreatorArcData.py	Python File	1 KB
CreatorArcSet.py	Python File	1 KB
README.pdf	Brave HTML Document	271 KB
requirements.txt	Text Document	1 KB

Figure 1 The program directory

To run the agi-io_creator.py file you will need to set up an environment to run in. This can be a stand-alone virtual environment or one that you normally use. These instructions assume you'll set up a new virtual environment just for this application.

Unzip the archive and open the folder. You should see something like what is seen in Fig. 1. The README.pdf is this set of instructions (so we'll assume you've already unzipped the archive). The only file that you need to be concerned with is the agi-io_creator.py file. The other 2 (CreatorArcData.py and CreatorAcrSet.py are boilerplate code that the application uses for storing data while the program is running).

To create your virtual environment; (Note: you may use any method you are familiar with, and this assumes you have venv installed. You could also use Conda or any other python install procedure.)

1. Open a terminal in the Agi-io directory
2. Type ***python3 -m venv venv*** (this creates a venv directory)
3. Enable the environment: (depends on your OS)
    a. . venv/Scripts/activate # for Windows
    b. . venv/bin/activate # for Linux/Mac
4. Install the requirements into the virtual environment
    a. pip install -r requirements.txt

Once that finishes you should now have everything setup to run the program. In the same terminal type: ***python agi-io_creator.py***

That's it. The application should run and you should see something very close to Fig. 2.

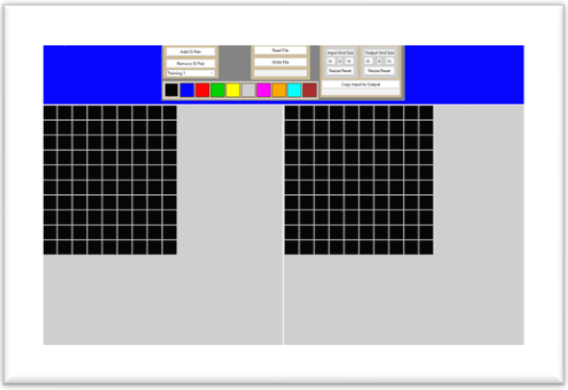

Figure 2 The initial application window.

Basic App Operations:

The application window is broken up into a few parts. The Upper section of the application window is where you can Read and Write files, add/remove additional IO pairs to the current working set, change the size of the Input and Output Grids and select colors from the color pallet.

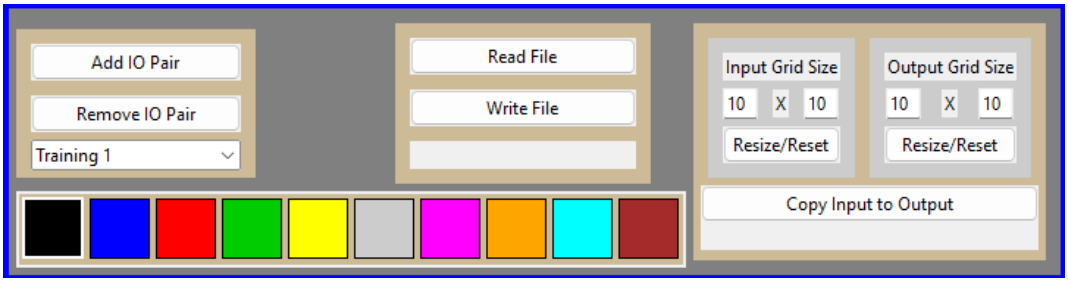

Figure 3 The tool pallet

The bottom section of the application is the Input and Output Grid where problems will be created.

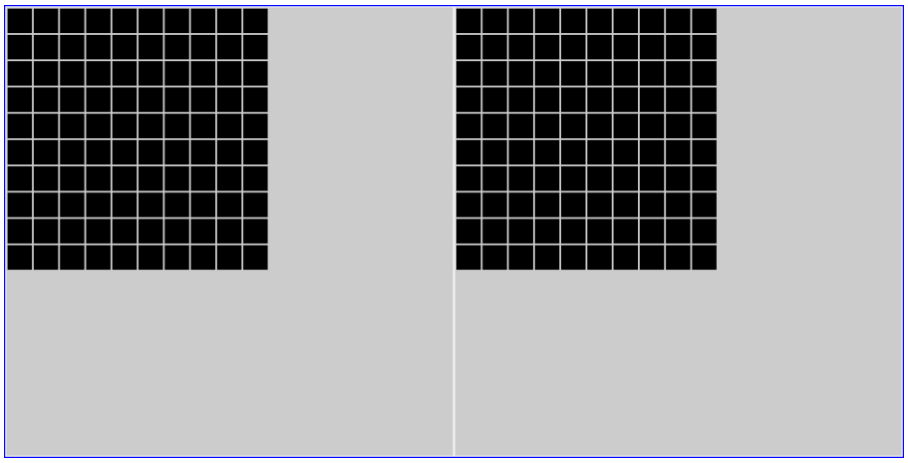

Figure 4 The Input & Output Grid

Basic Operations

When the application first opens there is 1 Training set, and 1 Test set included. Clicking on the drop-down combo box will allow you to switch between different input/output sets. Pressing the Add/Remove IO Pair buttons will add additional training sets or remove the training set (You can't add or remove test sets ~ but that will be a feature for future updates).

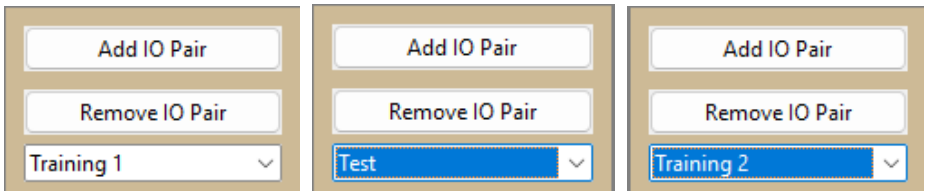

Figure 5 Adding/Removing Training Pairs

To resize the input and output grids you use the controls on the right side of the tool pallet. Each grid is controlled by the corresponding inputs, and each is independent of each other. Grids can be as small as 1x1 or as large as 30x30. When you change the size of each of the Entry boxes and push the Resize/Reset buttons it will change the corresponding grid. CAUTION and data you've created will be RESET!

The Copy Input to Output will resize the output grid and copy any data entered into the INPUT grid and duplicate it in the OUTPUT grid (including the Grid Size).

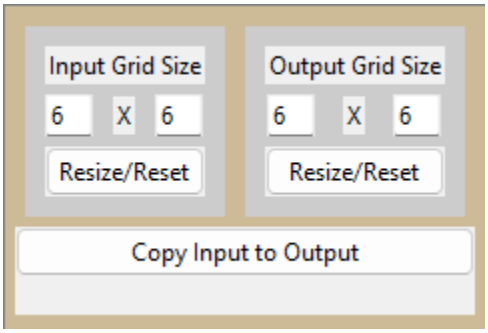

Figure 6 The Grid Input/Output controls

Creating Data

You create an Input or Output diagram by first selecting a color in the Color Pallet. The currently selected color will have a white outline around it. Fig. 7 shown below has Green selected. You may directly click on the color pallet to select a color or use the number keys 0-9 to select a color. You may use either the numbers on the keyboard or on a numeric keypad.

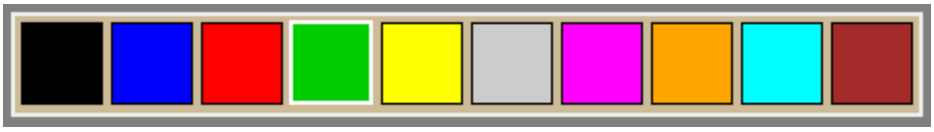

Figure 7 The Color Pallet

Once you have selected a color, you can click inside the input or output grid and that color will be displayed in the corresponding grid section that you selected.

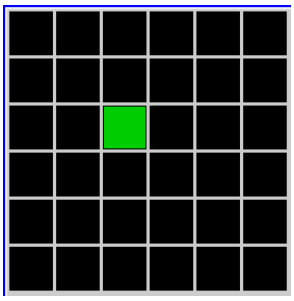

Figure 8 The Input Grid with Green inserted

You can continue to select colors from the pallet and click in either of the grids to insert those colors to create patterns. If you change the Set using the combo-box, the current grids will be saved internally, and you can come back to them later. Changing to another grid will load any existing data and when you return to the training/test set it will load any previous changes you have made.

Filling multiple grid locations at once.

You may fill multiple grid locations with the currently selected color by using the Ctrl Key with the mouse Left Click and drag. This will display a small blue box over the grid cells you are selecting. When you release the mouse button the cells will be filled with the currently selected color. This will aid in quickly creating patterns in the grid.

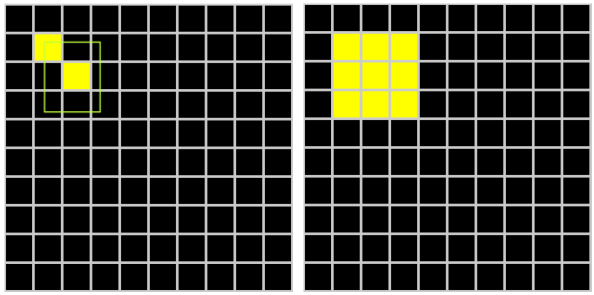

Figure 9 Filling multiple grid locations with the currently selected color.

Copy, Pasting & Deleting Data

To copy and paste or delete cells you can achieve this by using the Shift Key and the mouse. You must first highlight the grid cells. You do so by holding the Shift Key and left click & drag on the grid with the mouse which creates a Lt. Blue box. When you release the mouse button, all the grid cell locations under the selection box will highlight with an Lt. Blue outline. Once you have selected cells you can *Copy* or *Delete* the data selected.

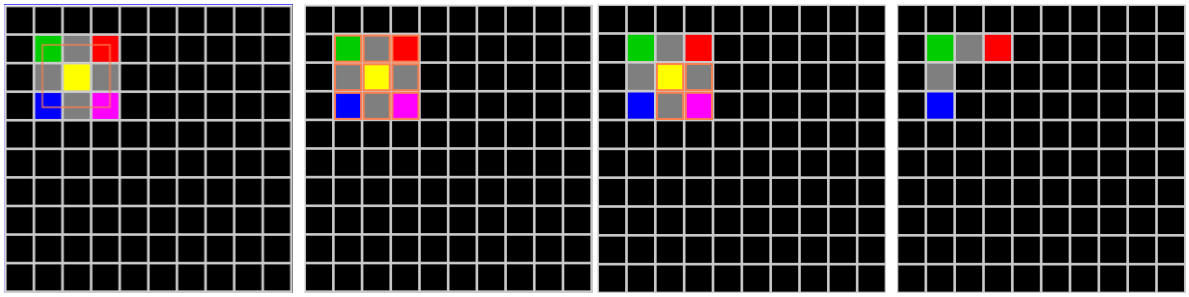

Figure 10 Select and Delete Cells

To delete data, highlight part of the grid and press the *Delete* button. This will remove the colors in the grid that are highlighted and reset them to empty (i.e. fill them with black) as well as removing the grid selection. You can also achieve this by using the multi select fill with the color pallet selection on the Black color.

To copy the highlighted data, use Control + C. This will copy the data in the cells to an internal clipboard data structure (and unhighlight the grid selection). To paste the data in a new location, you must first highlight a new grid location then paste with Control + V and a duplicate of the copied cells will be placed with the upper left corner of the copied cells filling in at the new highlight location as can be seen in Figure 11. You can then continue to highlight cells and paste the data as many times as you need to. You may also paste data between the input and output grids as there is no reliance where the source data is copied from.

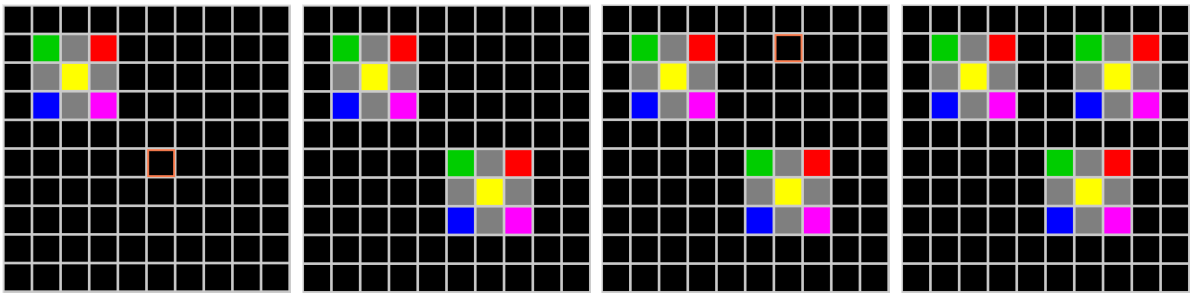

Figure 11 Copy and Pasting Data

If you highlight a cell and try to paste more data than there is available grid space, the data will be truncated to the available grid space as can be seen in Figure 12.

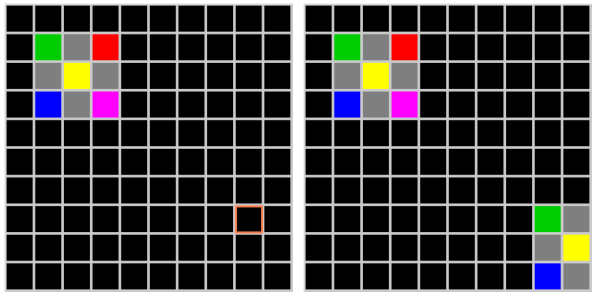

Figure 12 Pasting and truncation of data.

If you highlight cells that already contain data and paste into that location any previous data will be overwritten with the data contained in the clipboard as seen here in Figure 13. If you highlight more than one cell, it will use the topmost left cell in the highlighted group as the upper left cell of the pasted data.

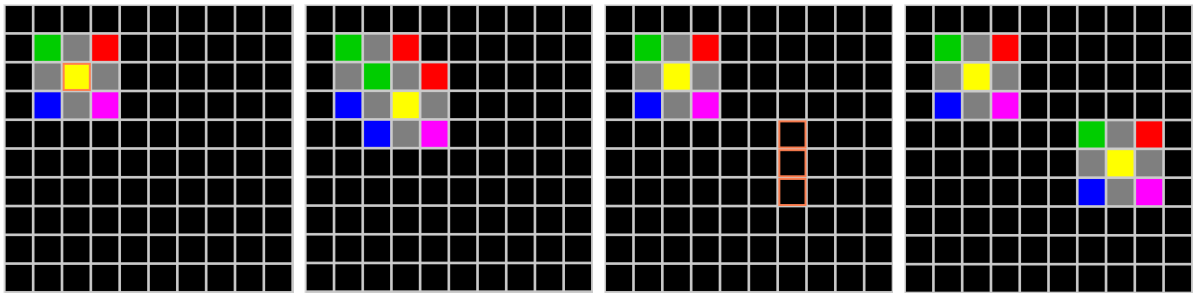

Figure 13

Special Paste

Special Paste can be accomplished using the CTRL-B key command which will not overwrite destination cells if the source cell is empty (i.e. filled with Black), allowing you to copy in blocks into existing objects essentially meshing them together. As can be seen here in Fig. 14 selecting the red blocks contain 2 empty cells. When using the normal CTRL-V to paste using the lower left blue block as target selection starting cell, the 2 blue cells get overwritten by the source blocks. Using CRTL-B to perform a special paste, the 2 blue cells are not overwritten, and the 2 shapes are merged.

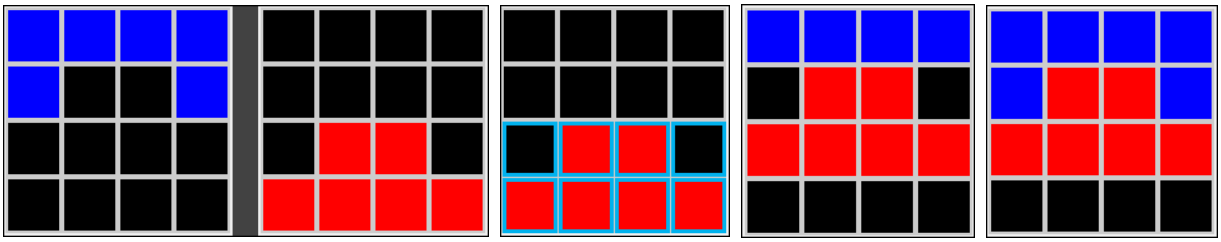

Figure 14 Using regular paste (CTRL-V) vs special paste (CTRL-B)

Rotating and Flipping Data

Flipping Data

You can highlight data and perform a horizontal or vertical flip of the data. Start by highlighting the data (Shift + Mouse Click & Drag) to select the data, then use ALT+h for a horizontal flip (Fig. 14 center) or ALT+v for a vertical flip (Fig. 14 right). When you perform a flip, the highlighted cells will remain highlighted until you either perform a highlight over another part of the grid or hit the ESCAPE key.

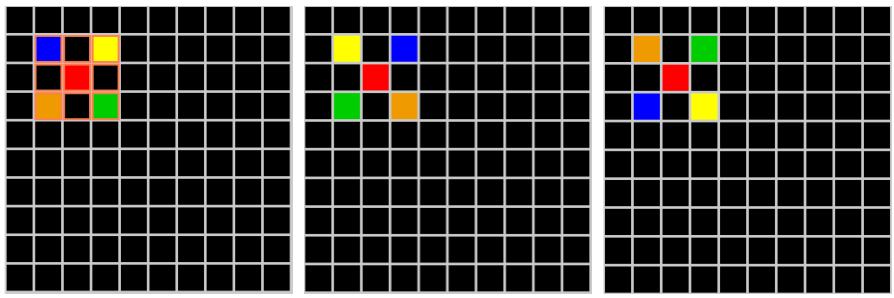

Figure 15 Flipping data horizontally (center) or vertically (right) from the starting position (left).

Rotating Data

Symmetrical Data

Highlight a selection and use the ALT key with the 7, 8 or 9 key. ALT + 7 rotates the selection 90 degrees CCW (or essentially 270 degrees CW) while ALT + 9 rotates the selection 90 degrees CW. Using ALT+8 rotates the selection 180 degrees (essentially a horizontal and vertical flip). After the rotation is performed, the highlight will still be active, allowing additional transformations to be made. To remove the highlight, press the ESCAPE button or highlight another part of the grid.

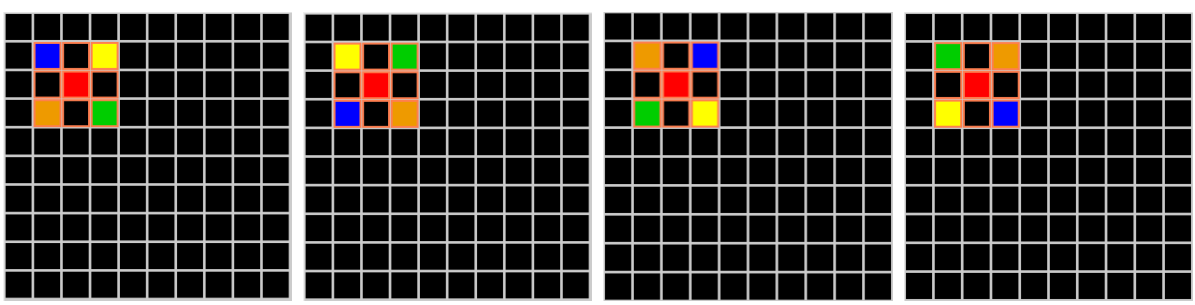

Figure 16 Starting position (left) highlights the selection.

press ALT+7 for 90°CCW (center-left), press ALT+9 for 90°CW (center-right), press ATL+8 for 180°rotation (right)

Non-Symmetrical Data

For non-symmetrical data (such as 3x4 or 8x10 etc.) rotation is performed and the grid is rotated using the upper left corner of the selected object as the anchor point. The grid is rotated to match the selection as can be seen below (Figure 16 left). It should be noted that if the rotated selection runs off end of the grid, then the data is truncated and no longer available in the selection (Figure 16 right).

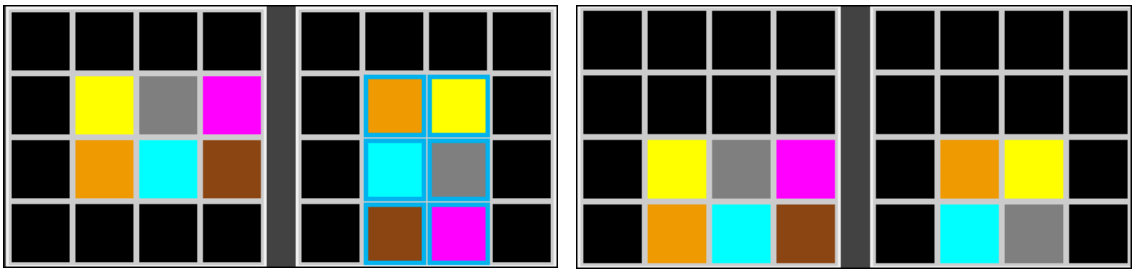

Figure 17 Rotation of a Non-Symmetrical Selection

Loading Data

To load preexisting data or data you may have created click on the Read File button which will open a dialog box that will allow you to navigate to the appropriate folder to load a file. When the file is loaded each of its Training sets and its Test set will be loaded and reflected in the combo-box.

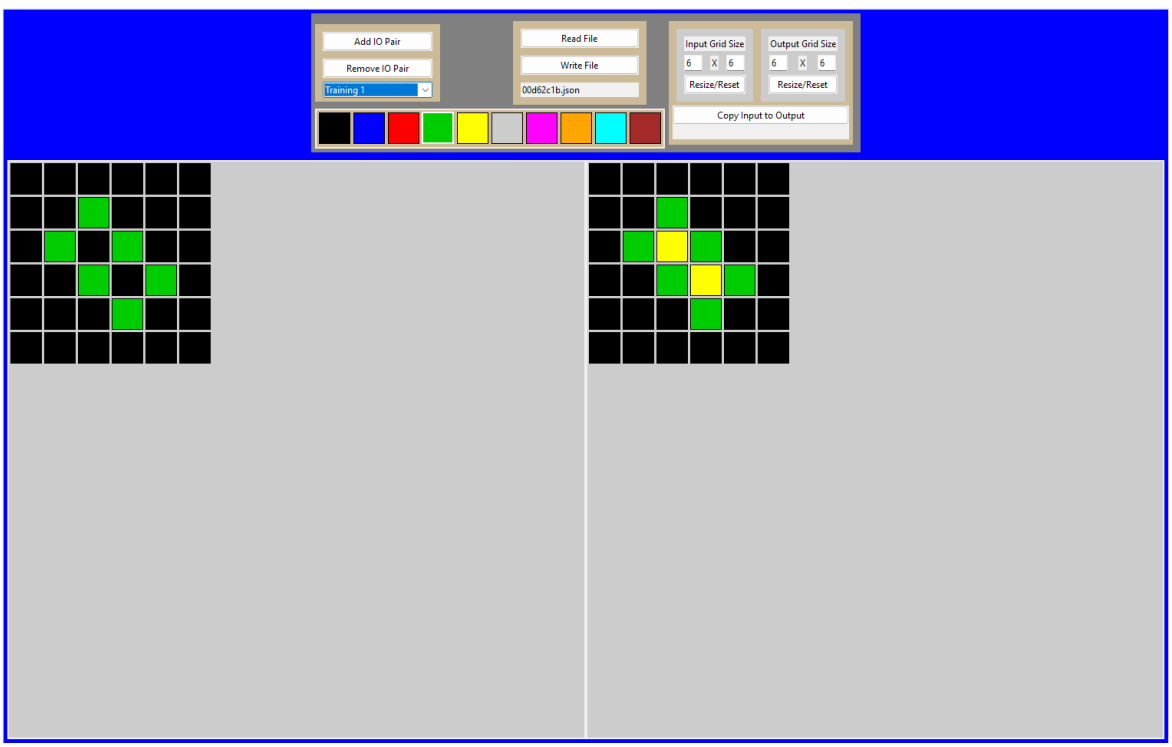

Figure 18 Loading a file

After you make any changes to an existing file, you should be sure to save them off before loading a new file or any existing changes will be overwritten. There is no warning when reading in new data.

Saving Data

To save your created data, you use the Write File button to open a dialog box. The box will already include ***a uniquely created name based on the data in the grids***. You can override this name if you choose to, but you then will have to assure that the file name you select is unique to any other problem, so it's suggested that you just use the existing file name.

Select a directory for saving the file in and click the Save button. The data will be written out to a Json file with the extension of '*.json*'.